Bachelor's Thesis — Final Project Document

Limits to Arbitrage in the Indian Equity Market

Why do mispricings survive, and what does that tell us about who sets prices?

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1. Overview and goal

This project studies why price errors survive in Indian stocks. Textbook theory says that if a stock is mispriced, arbitrageurs will trade against the error and push the price back to fair value almost instantly. In reality, mispricings persist for days or weeks. The standard explanation is that arbitrage is not free: the traders who would correct an error face capital limits, trading costs, and rules that cap how large a position they may hold. These frictions are the "limits to arbitrage."

India is an unusually clean laboratory for measuring these limits because they are unusually visible and, in several cases, mechanically imposed by the regulator. There is no genuine short selling in the cash market, the stock-lending-and-borrowing (SLB) market is thin, and there is an explicit rule — the F&O "ban period" — that halts fresh derivative positions in a stock once open interest crosses 95% of that stock's Market-Wide Position Limit (MWPL). The ban is mechanical, precisely dated, and published every day by NSE.

The goal is to build a validated, reproducible measure of how tightly arbitrage constraints bind across Indian stocks over time, and then to establish *causally* how much of the observed mispricing in Indian equities those constraints explain — using the MWPL/ban mechanism and the 2024–26 SEBI reforms as sources of identifying variation. The end product is both a research contribution and a public-data toolkit that other researchers can reuse.

2. Motivation and basis

2.1 Why this question matters

Whether a return pattern reflects risk or mispricing is one of the central questions in asset pricing. If a pattern earns higher returns precisely where arbitrage is hardest, that is evidence it is mispricing being prevented from correcting, not compensation for risk. Measuring the limits directly — rather than assuming them — lets us adjudicate this for the Indian market, where the evidence is thin. The question is also directly policy-relevant: through 2024–26 SEBI has been actively resetting position limits, explicitly trading off speculation control against market quality. This project measures the market-quality side of that trade-off.

2.2 Why India, and why now

Two features make the timing ideal. First, India's constraint architecture is distinct from the US and even from other emerging markets — the MWPL ban is a repeated, stock-specific, mechanically-triggered contraction in arbitrage capacity that has essentially no direct counterpart in the literature. Second, the 2024–26 reform wave delivered a dense sequence of dated, market-wide changes to exactly the constraints this project studies. These are natural experiments in which efficiency is the outcome variable, not the assumption. The key reforms are summarised below.

Reform	Effective	What changed
Minimum contract size raised to ₹15–20 lakh	20 Nov 2024 (phased to Feb 2025)	Fewer, larger index contracts; sharp reduction in small-retail participation
Weekly expiries cut to one benchmark per exchange	20 Nov 2024	Bank Nifty, Fin Nifty, Midcap Select weeklies removed; only Nifty 50 weekly retained on NSE
2% ELM on expiry-day short options; upfront option-premium collection	20 Nov 2024 (ELM) / 1 Feb 2025 (premium)	Higher cost of carrying speculative option positions

Reform	Effective	What changed
Calendar-spread margin benefit removed on expiry day	1 Feb 2025	Higher margin for expiry-day spread positions
Intraday monitoring of index position limits	1 Apr 2025	Limits checked through the day, not only end-of-day
New index-derivative position limits (net ₹1,500 cr / gross ₹10,000 cr per PAN); FutEq (delta-based) OI in force	1 Jul 2025	Tighter caps on large index positions; OI now measured in future-equivalent (delta) terms
New MWPL formula + ban-period rule recast in delta terms	1 Oct 2025	MWPL = lower of {15% of free float, 65× Average Daily Delivery Value (ADDV)}, floor 10% of free float, recomputed quarterly; during a ban, positions must reduce end-of-day FutEq (delta) OI
Intraday MWPL monitoring for single stocks (≥4 random checks/day); ≥14 constituents for non-benchmark index derivatives	3 Nov 2025	Stricter, higher-frequency MWPL enforcement
Pre-open session extended to F&O; index-option limits move to normal (delta) enforcement	6 Dec 2025	Further change to how the constraint binds

Crucially, from 1 October 2025 the MWPL denominator is tied to Average Daily Delivery Value — a trailing cash-market delivery quantity recomputed only once a quarter (rolling three-month ADDV) — subject to a 10%-of-free-float floor. Because that denominator moves for reasons unrelated to today's derivative speculation and is reset only quarterly, it is a clean source of variation in ban status, and it is central to the identification strategy in Section 7. SEBI states one explicit aim of this reform is to "reduce instances of spurious F&O ban periods in single stocks" — the very mechanism this project studies.

3. Research questions

Primary question. How strongly do arbitrage constraints bind in Indian equities, and how much of the observed, persistent mispricing can be attributed to them?

This decomposes into five sub-questions, each of which maps to a chapter:

1. Can we construct a single, credible, publicly-reproducible index of arbitrage-constraint intensity at the stock-day level?
2. Do mispricings (futures-basis deviations, put-call-parity violations, post-earnings drift) widen and persist longer when constraints are tighter?
3. Is the relationship causal — i.e., does exogenously-induced ban status widen mispricing, once we strip out the endogenous crowding that triggers bans in the first place?
4. How wide is the "no-arbitrage band" inside which prices do not correct, how does it vary across stocks, and did the 2025 reforms move it?

5. Cross-sectionally, do well-known return signals pay more in high-constraint stocks — the signature of mispricing rather than risk?

4. Hypotheses

Each hypothesis is stated with the theory it tests, so that a rejection is as informative as a confirmation.

#	Hypothesis	Grounded in
H1	Mispricing magnitude rises with the constraint index; deviations are both larger and slower to correct when constraints are tight.	Shleifer–Vishny (1997); Gromb–Vayanos
H2	Post-earnings-announcement drift is larger for constrained stocks, and largest on days a stock is actually in the ban.	Limits-to-arbitrage view of PEAD
H3	The no-arbitrage band is wider for smaller, less liquid, more retail-owned stocks, and widens with the constraint index.	Transaction-cost band models; Aggarwal (2015)
H4	The Oct 2025 (delivery-value-linked MWPL, delta-based ban rule) and Dec 2025 changes narrow the band and shorten mispricing half-lives.	Constraint-intensity interpretation of the reforms
H5	Known return signals earn higher returns among high-constraint stocks, consistent with mispricing rather than risk.	Stambaugh–Yu–Yuan (2015)
H6	Mispricing responds to ban status induced by MWPL-denominator revisions (variation orthogonal to current crowding), not only to endogenous crowding.	Identification / falsification

5. Conceptual framework

The organising idea is the no-arbitrage band. In a frictionless market, the futures price equals its cost-of-carry fair value and put-call parity holds exactly; any deviation is instantly traded away. With frictions — financing costs, short-sale and lending limits, and hard position caps — there is a band around fair value inside which the expected profit from an arbitrage trade does not cover its cost, so prices are left uncorrected. Only when a deviation pushes past the edge of the band does correction kick in. The width of this band is a direct, interpretable measure of the limits to arbitrage.

This yields a clean chain of reasoning that the whole project tests: tighter constraints → a wider band → larger and more persistent mispricing → and, in the cross-section, higher realised returns to signals that bet against those mispricings. The MWPL ban is a sharp, observable tightening of the constraint; the 2025 reforms are dated shifts in its intensity. The band is naturally a non-linear, threshold object — prices barely adjust inside it and adjust quickly outside it — which is why the estimation uses threshold and smooth-transition models rather than linear regressions, and why Prof. Chaubal's work in non-linear time series is a strong fit.

6. Literature and the research gap

The project sits on three bodies of work. The theory of limits to arbitrage — Shleifer and Vishny (1997), the Gromb–Vayanos survey on capital-constrained arbitrage, and Ofek, Richardson and Whitelaw (2004) on put-call-parity violations under short-sale constraints — provides the mechanisms and the mispricing measures. The asset-pricing consequence is Stambaugh, Yu and Yuan (2015), whose test of whether anomalies

concentrate where arbitrage is hardest is the template for the cross-sectional chapter. The econometrics of transaction-cost bands — threshold and smooth-transition error-correction models of index-futures arbitrage — provides the estimation toolkit.

The gap. The closest Indian precedent is Aggarwal (2015, IGIDR), who measures no-arbitrage bands between single-stock futures and spot prices and finds funding liquidity drives their common variation — but she does not use the F&O ban / MWPL mechanism at all. Internationally, position limits have been studied in Korea and Taiwan, but India's specific MWPL ban is essentially unstudied as a shock to arbitrage capacity. The existing Indian literature on derivatives and mispricing is largely descriptive. So the measurement method exists and is validated, and the constraint mechanism exists and is precisely observable — but no one has tied the two together. That junction is the contribution.

7. Methodology

The work is organised into five methodological blocks that also serve as thesis chapters, plus a concluding policy chapter. Each block uses a genuinely different technique, which gives the thesis the breadth expected at Stage II.

7.1 Constructing the arbitrage-constraint index (Chapter 1)

For each stock and each trading day, assemble a panel of constraint proxies and reduce them to one index using principal-component analysis (with a latent-factor model as a robustness alternative). Candidate components:

- Ban status (in-ban indicator) and length of the current ban spell.
- MWPL utilisation — open interest as a percentage of the stock's MWPL.
- Futures-basis dispersion and magnitude (how far, and how variably, the basis strays from cost-of-carry).
- Idiosyncratic volatility — residual volatility from a factor model, the classic arbitrage-risk proxy.
- Delivery percentage (from the MTO/full bhavcopy) as an inverse proxy for speculative, non-delivery trading.
- Ownership breadth from quarterly shareholding filings — investor-category shares and holder concentration.
- Illiquidity (Amihud measure, bid-ask proxy) and SLB availability/cost where obtainable.

This chapter is deliberately designed to be a stand-alone deliverable: a validated, reproducible measure of arbitrage-constraint intensity for Indian equities, built entirely from public data, is publishable and citable even if every later causal test comes back null.

7.2 Measuring mispricing (Chapter 2)

Three complementary, well-understood measures, so that no single noisy measure drives the results: (i) deviation of the futures price from its cost-of-carry fair value; (ii) put-call-parity deviations, restricted to near-the-money options because settlement prices for thinly-traded strikes are unreliable; and (iii) the magnitude of post-earnings-announcement drift around scheduled results. Each is computed at the stock-day (or stock-event) level so it can be matched to the constraint index.

7.3 Establishing causality (Chapter 3)

The central threat is reverse causality: a stock enters the ban precisely because speculative interest crowded into it, which is also when mispricing is most likely. A naive "ban → wider basis" regression would confuse the two. The identification strategy exploits the way the ban threshold (95% of MWPL) is set. Before the reform MWPL was 20% of free float (or 30× the prior month's average trading quantity); from 1 October 2025 it is the lower of 15% of free float or 65× the trailing Average Daily Delivery Value (ADDV), floored at 10% of free float and recomputed only once a quarter on a rolling three-month window. In both regimes the denominator moves for reasons unconnected to today's derivative trading — free float changes with issuance and shareholding shifts, and post-reform the binding term is a quarterly-reset delivery-value figure — so the same open interest can be in-ban or not depending on stale, exogenous inputs. These denominator revisions give variation in ban status that is plausibly orthogonal to current mispricing, and serve as the instrument (the post-2025 ADDV rule, being delivery-based and quarterly, is the cleaner instrument). A regression discontinuity around the 95% threshold is reported as supporting evidence, accompanied by a McCrary density test for manipulation/bunching (the running variable is anticipated, so some bunching is expected — which is exactly why the RD is supporting, not primary).

7.4 The no-arbitrage band and the 2025 regime shifts (Chapter 4)

Model price adjustment as a threshold process: little correction inside the band, fast correction outside it. The workhorse estimator is a three-regime threshold vector error-correction model (TVECM) with two asymmetric thresholds and an unknown cointegrating vector (Ters and Urban, 2020, extending Hansen and Seo, 2002), which recovers both the no-arbitrage band and the implied overall arbitrage cost and is designed for exactly this kind of pre/post-event analysis; a three-regime TAR (following Aggarwal, 2015) and a smooth-transition (STAR) specification are run as robustness. The band width is allowed to depend on the constraint index from Chapter 1. The dated 2025 reforms then serve as market-wide shifts in constraint intensity — the FutEq (delta-based) redefinition of open interest (in force by mid-2025), the new delivery-value-linked MWPL and the delta-based ban-period rule (1 October 2025), and intraday MWPL monitoring (3 November 2025) — analysed with pre/post event windows and, where a cross-section of treatment timing exists, staggered-treatment-robust difference-in-differences estimators (Callaway–Sant'Anna, Sun–Abraham).

7.5 Cross-sectional asset-pricing test (Chapter 5)

Using Fama–MacBeth regressions, test whether well-known return signals earn higher returns among high-constraint stocks — the direct Stambaugh–Yu–Yuan test, which also reuses the existing 45-signal long-short infrastructure. Because the derivatives universe is only ~220 stocks, statistical power is limited; this chapter is reported as secondary, corroborating evidence with honest power caveats, never as the load-bearing result.

7.6 Policy discussion (concluding chapter)

The concluding chapter borrows the framing of the retail-participation debate — retail crowding is one reason stocks hit the position limit in the first place — and reads the results against SEBI's stated trade-off between curbing speculation and preserving market quality. This delivers the policy relevance of the alternative retail project without depending on the firm-level retail-trading data that public sources cannot provide.

8. Data sources and access

Every dataset is free and public, sourced from the exchange. This is a deliberate design choice: it keeps the project fully reproducible and removes any dependence on paid or proprietary feeds.

Dataset (file)	Source	Key contents	Frequency
Cash bhavcopy — sec_bhavdata_full	NSE	OHLC, volume, and delivery quantity & % (delivery value → Average Daily Delivery Value, the post-2025 MWPL input)	Daily
Cash bhavcopy — UDiFF (BhavCopy_NSE_CM…)	NSE	ISIN-keyed OHLC (current file format from 2024)	Daily
F&O bhavcopy (fo…bhav / BhavCopy_NSE_FO…)	NSE	Futures & options settlement prices, open interest, volume, contract specs	Daily
Combined OI / MWPL file	NSE	Per-stock MWPL, open interest (notional and, post-2025, FutEq/delta), and OI as % of MWPL	Daily
Securities in ban period	NSE	List of stocks in F&O ban that day	Daily
Quarterly shareholding pattern	NSE/BSE filings	Ownership by category (promoter, FII, DII, individuals ≤₹2 lakh)	Quarterly
Corporate results / announcements calendar	NSE	Earnings dates & timestamps (for the PEAD measure)	Event
India VIX	NSE	Implied-volatility index	Daily
Index constituents & F&O eligibility lists	NSE	Derivatives-universe membership over time	Periodic

Access is by direct download of the daily report files from NSE's archives (with well-maintained open-source wrappers such as the R package nser and Python scrapers available to automate bulk retrieval). Archives extend comfortably past the proposed sample start of 2015. A note on the mid-sample format change: NSE migrated its bhavcopy files to the ISIN-keyed UDiFF format in 2024, so the data pipeline must handle both the legacy and current layouts — a routine but necessary cleaning step.

8.1 Sample and construction

Proposed sample: 2015 to 2026, covering the roughly 220 stocks in the F&O segment (with membership tracked over time, since the eligibility list changes). Pre-2019 data can serve as a placebo/pre-retail-era period. Construction proceeds from raw daily files to a cleaned stock-day panel: merge cash and F&O records by symbol and date, compute the basis and cost-of-carry fair value, attach ban status and MWPL utilisation, join quarterly shareholding (forward-filled within quarter), and flag earnings-event windows. Two construction points follow from the reform: (i) MWPL must be reconstructed under both regimes — 20% of free float (or 30× prior-month volume) before the change and, from 1 October 2025, the lower of 15% of free float or 65× rolling-three-month ADDV with a 10%-free-float floor — and switched at the change date, which also yields the denominator revisions used as the instrument; (ii) the post-2025 FutEq/delta open-interest measure requires option deltas, computed from settlement-price-implied volatilities. All cleaning steps are scripted and version-controlled so the dataset is reproducible end-to-end.

9. Expected results

- Mispricing measures widen and correct more slowly when the constraint index is high, with the effect strongest on days a stock is actually in the ban.
- Post-earnings drift is larger in constrained stocks, since the trades that would close the gap are the ones being blocked.
- The estimated no-arbitrage band is wider for smaller, less liquid, more retail-owned stocks.
- The October and December 2025 relaxations show up as a narrowing of the band and shorter mispricing half-lives.
- Return signals pay more in high-constraint stocks, supporting a mispricing rather than a risk interpretation.

10. Deliverables and robustness to a null result

The single most important design principle is that Chapter 1 — the validated, reproducible arbitrage-constraint measure — is a contribution in its own right. Theses fail when their entire value hinges on the sign of one coefficient; this one is built so that even if every causal test is insignificant, the constraint index, the cleaned public-data panel, and the descriptive validation remain a usable, citable asset. Concrete deliverables: (i) the reproducible stock-day dataset and cleaning code; (ii) the constraint index and its documentation; (iii) the causal and time-series results; and (iv) the written thesis.

11. Risks and mitigations

Risk	Mitigation
Reverse causality — bans are triggered by the same crowding that causes mispricing.	Instrument ban status with MWPL-denominator revisions; treat the RD as supporting evidence only.
RD weakness — the 95% running variable is anticipated and may bunch.	Run the McCrary density test in the first month; lean on the instrument and dated reforms if bunching is severe.
Low cross-sectional power — only ~220 F&O stocks.	Report the Fama–MacBeth chapter as secondary, with explicit power caveats.
Unreliable option prices for thin strikes.	Restrict put-call-parity tests to near-the-money contracts; rely on the futures basis as the primary mispricing measure.
Mechanism changed mid-sample (Oct & Dec 2025).	Split the sample at the change dates and treat each change as an additional, dated constraint-intensity shock.
Bhavcopy format change (UDiFF, 2024).	Build the pipeline to parse both legacy and current file layouts.

12. Indicative timeline

Stage I — EC 495 (Semester 7)

- Weeks 1–3: literature review; finalise the mispricing measures and the constraint-index component list.

- Weeks 4–8: build and clean the full stock-day dataset from NSE files.
- Weeks 9–12: construct the constraint index; descriptive validation; run the McCrary test and an instrument-strength check to confirm the identification is viable.
- End of semester: interim report with dataset, constraint measure, and preliminary findings.

Stage II — EC 497 (Semester 8)

- Weeks 1–5: main causal estimation (IV) and the 2025 rule-change analysis.
- Weeks 6–9: threshold/smooth-transition band models and the cross-sectional Fama–MacBeth tests.
- Weeks 10–12: robustness checks, policy chapter, and thesis writing.

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